“CELLULAR PARABIOSIS” WITH MORIBUND “FEEDER LAYER” CELLS?

Miro for Sara: modified from the letter of 10/12/18 in Split:

Dear Sara,
I'll repeat how I understood the details of your and Mila's version of the Newbold experiment:
100 6-TG resistant V79 cells are mixed with 750.000 of V79 TG-sensitive cells in a "zero time" - then, left for 4 hours to attach (and do whatever they do) - then either 6-TG alone, or 6-TG and TPA/PMA together, are added - followed by incubation for 72 hours when the cells are fixed and stained by crystal violet to observe surviving cells/colonies.
This means that parabiosis of TG-resistant cells with living TG-sensitive cells can occur only in an unknown fraction of the 4 hour period! After these 4 hours, TG-sensitive cells are killed (or moribund, or "senescent"?) by TG and the TG-resistant ones try to grow among the moribund and dying TG-sensitive V79 cells - but most are phenotypically suppressed and die. This situation reminds “feeder layer” cells (which are previously killed by radiation or mitomycin C) except that: parabiosis will allow particular cells to grow on feeder layer while in our Newbold expt. parabiosis will allow killing of TG-resistant cells by TG!
~~Now, it is not conceivable that growth of an entire TG-resistant colony will be phenotypically suppressed because of an initial - less than 4 hour - period of potential contact.~~ (*My wrong thinking: phenotypic suppression by parabiosis will kill the initial TG-resistant cell and that’s enough not to make ever a colony!*) ~~Therefore, the only way to get the Newbold result is that **there is parabiosis between dying TG-sensitive cells and living TG-resistant cells** - as to make resistant cells phenotypically sensitive. Such necrophilic metabolic crosstalk between excess of killed cells and a small number of live cells -- known since ages as "feeder layer technique" to grow cultured cells that cannot grow alone (ES cells, for example).~~ There is also the famous "bystander effect" (read about it!).
Both soluble factors (conditioned medium) and direct contact are needed - like for parabiosis of malignant phenotype (G. Klein). Please,

read carefully the attached recent paper on feeder layer cells (they are also sold by Sigma).

~~This interpretation would solve our problem - cellular parabiosis in Newbold expt. is perhaps a **"metabolic cannibalism" - a version of cellular parabiosis?** Best feeder layers are from fibroblasts; best phenotypic suppressors of malignant growth are fibroblasts, etc...~~ This is my bullshitting! The question is simply: **Does feeder layer allow cell growth by cellular parabiosis?** If it does, then TPA/PMA should prevent it and - we have a simple routine test for cellular parabiosis! Sara, unless somebody in your lab uses feeder layer to grow particular cells, Vlatka could do this test in Niels' lab when growing ES cells on feeder layers (simply add TPA/PMA to one parallel plate with feeder layer and see whether ES cells grow!).

I want you to talk with colleagues, read about this bizarre practice of feeder layers, etc. - that is still the only way to grow many cell types (iPS and ES cells par excellence): you would expect this from cellular parabiosis!! It would be great to understand this old feeder layer fetishism!! Think, talk, read - it's never too much! Don't leave the fun to others. We all need this activity to be productive scientists.

From Miro, in Paris 11 dec. 2018:

<u>Questions to answer experimentally - first phenomenologically, then mechanistically:</u>

(A) Are dead cells better or worst parabiotic partners than live cells?
Is cellular parabiosis a "*samaritan*" help to sick cells (to make them healthy by cellular "solidarity") **or**:
do senescent/sick/dying cells serve a tissue *"capitalism" to make healthy cells even healthier* at expense of sick and poor serving as sacrificed parabiotic source of "food" for living cells, i.e., some kind of biosynthetic cannibalism? Ideologically more positive is this: moribund cells will die anyhow - let's make them useful to the community! In a nutshell, are dying, or even dead, cells even better parabiotic partner than healthy

living cell? May be that dying cells grow more TnTs? Unlikely, but go for a surprise (expt 1).

(B) Will delayed addition of TPA/PMA (expt. 3a) or delayed addition of 6-TG (expt. 3b) tell us something interesting when compared with data from expt. 2.

EXPERIMENTS:

(1) Irradiate (or treat with mitomycin C) the V79 $Hprt^+$ cells (as in the literature on preparing feeder layers) before depositing on plates (+/- glutaraldehyde crosslinking to plates) - or buy commercial (Sigma Aldrich) feeder layer kit.

Add 100 $Hprt^-$ cells to 7.5 x 10^5 of these irradiated deadly damaged $Hprt^+$ cells and do your usual Newbold expt. by addition of 6-TG. See, what result do you get.

(2) Follow in time the dying of V79 $Hprt^+$ cells <u>alone</u>: add simultaneously at "zero time" (4 h?) 6-TG to 10 plates seeded with equal number of V79 $Hprt^+$ cells; incubate all plates and withdraw one-by-one at: 0, 2, 4, 6, 8, 12, 16, 32, 48, 54 h to stain with Alexa hydrazide or any similar staining for dead (membrane permeable) cells. You will know the kinetics of the "clinical death" of V79 $Hprt^+$ cells in 6-TG.

(3) (a) Mix, as usually, 100 $Hprt^-$ cells and 7.5 x 10^5 V79 $Hprt^+$ cells - but in 10 parallel plates - wait 4 h and add 6-TG to all 10 plates, and then - while incubating - add TPA/PMA (one-by-one plate!) at 4, 8, 12, 16, 20, 30, 40, 50, 60 and 70 h. Keep all plates incubating and fix all cells at 72 h and stain with crystal violet, as usually (Look for cells eventually detaching; if massive, we may need to fix cells to the plates by glutaraldehyde?).

(b) Mix 100 $Hprt^-$ cells with 7.5 x 10^5 V79 $Hprt^+$ cells and TPA/PMA in 6 parallel plates; then, while incubating, take out single plates and add 6-TG at 4, 6, 8, 10, 15, 25 h and continue incubating until 72 h - fix and stain with crystal violet.

Compare data from (2) with (3a) and (3b) and think!

There are certainly better experiments to answer these questions - <u>everybody is invited</u> to come up with better experiments than these! - but, beforehand talk, talk and think about. Don't keep your thoughts to yourself, because it is often unproductive (example is in my first letter).